Supplement

Gemma 4 Rubber-Stamps Its Own Failing Patches

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github.com/Vein05/gemma4-paper-track Paper: `rubber-stamp.pdf` Kaggle paper track

This supplement accompanies *Gemma 4 Rubber-Stamps Its Own Failing Patches* (`rubber-stamp.pdf`). Terms, patch sets and measures are defined there (Section 2). Every number traces to a file in `scripts/derived/` of the repository, https://github.com/Vein05/gemma4-paper-track.

A Setup details and judge prompt

Issues. The held-out issues are the first 200 of a fixed random order (seed 0) of SWE-bench Verified issues outside the Mini issues for which at least one of the five other writers has a usable patch (447 issues). Five pilot issues drawn first were left out of the sample. A usable patch is non-empty after removing test files and at most 30,000 characters long.

Agent. The 4-bit competition checkpoint `gemma-4-31B-it-qat-w4a16-ct` (revision `52f3f65b`) ran in mini-SWE-agent v1.7.0 with its SWE-bench configuration unchanged (250 steps, temperature 0, 60-second command timeout), served by vLLM with a 131,072-token context on one NVIDIA L40S. Docker was replaced by Apptainer sandboxes of the same SWE-bench images; grading used SWE-bench 4.1.0's grading code in the same kind of sandbox.

Judges. Outside judges for a model exclude its own family and, for each patch, the family of the patch's writer. Gemma as judge is the hosted model. OpenRouter routed its 1,064 held-out calls to nine providers. Most of them (983) went to providers that listed 4-bit or 8-bit weights on 2026-10-05. Of the rest, 79 went to providers that listed no precision and 2 to a provider that served bf16. We did not record the precision of each call (provider per call: `scripts/derived/fresh_gemma4-31b.jsonl`). In an earlier one-word check on 173 other patches, the hosted model pinned to a bf16 provider and the 4-bit checkpoint gave the same answer in 331 of 344 calls. The re-ask limit was 8,000 output tokens for our own patch sets and 3,000 for the mixed-harness set; other writers' patches on the Mini issues keep their mixed-harness scores.

Gemma as judge, served locally. We repeated the held-out judging with the competition checkpoint itself, served by vLLM 0.30.0 on one NVIDIA A100 at temperature 0 with thinking off, under a plan fixed before any score (`spec/gemma-31b-local-judge.md`). It gave a score to all 1,063 prompts on the first ask (rows: `scripts/derived/fresh_gemma4-31b-local.jsonl`).

On the paper's patches, the local judge gives 100 to 54% of Gemma's failing patches and to 32% of other writers'. The hosted model gives 61% and 35%. On failing patches, the local judge's leniency difference is +13.4 [6.9, 20.0]. For the hosted model it is +16.6 [10.3, 23.1]. The paired difference, hosted minus local, is +3.2 [−0.5, 7.2]. We detect no difference between the two judges, but the interval allows the hosted model to be up to 7.2 points more lenient. On passing patches, the local judge's leniency difference is +5.0 [0.8, 9.0]. For the hosted model it is +3.4 [−2.0, 8.0]. The local judge's AUROC is 0.658 on Gemma's patches and 0.762 on other writers'. For the hosted model they are 0.622 and 0.751.

Prompt. Every judge received the same system and user messages at temperature 0 (`scripts/selfjudge_matrix.py`, `scripts/selection_test.py`). Braces mark the parts filled in per patch.

```
System:  You are an expert Python maintainer.  You review a proposed fix for a reported issue.  You
cannot run code; judge from the material given.

User:  A user reported the issue below for the repository {owner/repo}.
<issue>{issue text}</issue>
A coding agent produced the patch below.
<patch>{patch without test files}</patch>
Question:  after this patch is applied, will the maintainers' hidden tests for this issue pass?  The
hidden tests were written for the upstream fix.  They check the behaviour the issue asks for, and tests
that passed before must still pass.  Explain briefly, then end your reply with a final line that is
exactly SCORE: <n>, where n is your probability from 0 to 100 that the hidden tests pass.
```

B Agent runs and grading

Agent runs. The 200 held-out issues ran as two jobs of 100. In the first, 98 runs submitted a patch, one ran out of the 131,072-token context and one stopped after repeated model-call timeouts; three patches were empty. In the second, one submitted patch was empty. On the Mini issues, 49 runs submitted and one stopped after repeated timeouts. Empty patches and runs that stopped count as unresolved, as in SWE-bench. One Mini issue (`django__django-11848`) failed to start its sandbox and was run again; the Mini analyses use the second run's patch.

Reference-patch check. We graded the official reference patch of every issue we ran (5 pilot, 200 held-out, 50 Mini). 248 of 255 resolve in our environment. The first pass labelled 16 matplotlib and xarray issues as timeouts; logs showed the container build, not the tests, exceeded a 1,200-second limit, and with a 3,600-second build limit all but three resolved. The 7 issues whose reference patch does not resolve were dropped from the held-out set (`astropy-7606`, `matplotlib-20826`, `-23314`, `-26113`, `-26208`, `requests-2931`, `sphinx-9711`); Gemma resolved none of them.

Agreement with official labels. Before using our grading, we graded 30 gpt-oss-120b patches with known official labels, stratified by repository. 26 of 30 matched. The four misses were environment failures (a network-dependent test, Ghostscript file access) and patches that apply only with fuzzy matching on today's rebuilt images. Other writers' patches therefore keep their official labels.

Resolve rates. Gemma resolved 56 and 42 of the two held-out batches of 100, 98 of the 193 kept issues in all, and 22 of the 50 Mini issues. On the Mini issues two sphinx patches ran past the 1,800-second test limit; counting them as failing, as SWE-bench does, changes the matched-harness gap from +0.066 to +0.081 [−0.010, 0.178].

C Re-asks and sensitivity

Table 1. Answers without a parsable score on our own patch sets (1,063 patches per judge), re-asked once with an 8,000-token limit. Most had run out of the 3,000-token limit before the score line.

Judge	Re-asked	Recovered	Left out
Gemma 4 31B	1	0	1
gpt-oss-20b	14	14	0
Qwen3.5-9B	74	73	1
GLM-4.7-Flash	52	42	10
Qwen3-Coder	0	0	0
gpt-oss-120b	5	0	5
GLM-4.5	1	1	0
Kimi-K2	129	117	12
Llama-4-Maverick	0	0	0

Sensitivity to the re-ask (exploratory). Leaving out patches whose answer needed a re-ask, the held-out self-judging gap is +0.030 [−0.024, 0.083] (all judges' re-asked patches; 763 patches, 143 of Gemma's), +0.055 [0.001, 0.106] (only Kimi-K2's) and +0.063 [0.012, 0.115] (only Qwen3.5-9B's). Without the re-asked patches, Gemma's own AUROC rises from 0.622 to 0.670, its leniency on failing patches is +11.1 [3.8, 18.2] and its rubber-stamp rate 56%.

Leniency on ranks. On per-judge percentile ranks instead of raw scores, the held-out leniency is +0.085 [0.048, 0.122] on failing patches and +0.024 [−0.007, 0.055] on passing patches.

D What the deficit is not

We checked four simple explanations on the mixed-harness patches, where Gemma has several attempts per issue.

Patch size. Gemma's patches are much smaller than the other writers' (7.7 against 72.7 added lines on average). Among single-file failing patches, matched by size in four bins, Gemma's leniency on its own patches stays 24.1 points [12.7, 39.9] above its leniency on other writers' patches (Table 3).

Missing reasoning. We rated every Mini patch again with Gemma's thinking switched on (Table 2). Its AUROC on its own patches does not move. It also gives 100 to the same share of its own failing patches: 48 of 79 with thinking off and 49 with it on, 40 of them in both. Thinking makes its other scores more extreme. The own failing patches it does not score 100 drop from a mean of 34 to 14 (−19.7 [−34.0, −4.5]). Its mean leniency on its own failing patches therefore falls from +12.9 to +6.2 points. On other writers' failing patches it rises from −8.1 to −5.5. The self-judging gap shrinks because thinking makes Gemma less accurate on other writers' patches.

Chen et al. (2025, cited in the paper) count how often a judge prefers its own wrong answer to another model's correct one. Our closest measure pairs Gemma's failing patch with another writer's passing patch on the same issue. There are only 52 such pairs. Gemma scores its own patch higher in 17% of them with thinking off and in 12% with it on, and the two tie in 42% and 46%.

Table 2. Gemma 4 31B with reasoning off and with thinking on, Mini issues, mixed-harness patches: the 331 patches that all nine judges scored and Gemma scored in both modes (two patches have no thinking score). Leniency is own minus other writers' failing patches, as in Section 3.1 of the paper. `derived/thinking_mode_2026-10-22.json`.

	Reasoning off	Thinking on
AUROC, own patches	0.638	0.638
AUROC, other writers' patches	0.836	0.795
Own failing patches scored 100	61%	62%
Own failing patches scored 95 or above	66%	66%
Own failing patches below 100, mean score	34	14
Other writers' failing patches scored 95 or above	22%	35%
Leniency [95%]	+21.0 [11.5, 30.3]	+11.7 [1.1, 22.3]
Self-judging gap [95%]	+0.138 [0.063, 0.214]	+0.097 [0.010, 0.192]

Issues it cannot solve. Gemma scores 66% of its failing patches at 95 or above on issues that one of its attempts solved (29 patches) and 67% on issues none solved (51). On other writers' failing patches for the unsolved issues, it is slightly stricter than outside judges (−5.8 points). So "a model cannot verify what it cannot solve" does not fit.

What Gemma writes when it rubber-stamps (exploratory). Gemma scored 52 of its own failing held-out patches 100. At least two outside judges scored 31 of these at 50 or less. We set this rule after a first look at the answers. For each case, a coder found the main defect, the reason the low-scoring outside judges give for their low score, and asked whether Gemma's answer names it (`data/hand-codes/`, with the codebook). Two coders coded every case on their own. Both were Claude models: Sonnet, then Opus without access to the first coder's codes. The first coded 5 cases as naming the defect and the second 6. They agree on 26 of 31 (Cohen's $\kappa = 0.45$): both find the defect never mentioned in 23 and named in 3. Two of the five split cases come from choosing the main defect when the outside judges name several. The other three turn on whether Gemma describing the flagged change, with no doubt about it, counts as naming the defect.

The codebook also asked whether Gemma states the opposite of the outside judges' claim. The two coders agree on this in only 13 of 31 cases ($\kappa = 0.03$), so we do not report it. One split case is django-15037. There Gemma calls `to_fields=[target_field]` "the correct way to specify a non-PK reference", and GLM-4.5 (score 15) answers that Django's keyword is `to_field`. Two other low-scoring judges give a leftover script in the patch as their reason, and Gemma mentions that script. `derived/rubber_stamp_reasons_2026-10-22.json`.

Choosing among attempts. We split the pass/fail pairs of Gemma's own patches into pairs from the same issue and pairs from different issues and computed pair accuracy, with ties counted as half. On 47 same-issue pairs Gemma scores 0.628 against 0.646 for the outside judges (−0.019 [−0.243, 0.148]). On 5,553 different-issue pairs it scores 0.636 against 0.706 (−0.070 [−0.138, −0.006]). Gemma gives exactly 100 to 110 of its 150 own patches, so its scores carry little information about which issue a patch belongs to; breaking those ties perfectly would raise its AUROC from 0.64 to 0.91.

We cannot yet tell whether Gemma recognizes its own patches or shares the mistakes in them. It is not more lenient toward other writers' failing patches that resemble its own wrong fix on the same issue (+5.6 [−5.5, 17.4]), but most of those patches share almost no changed lines with Gemma's, so this test is weak.

Table 3. Gemma's leniency on single-file failing patches by number of changed lines, mixed-harness Mini patches. The size-matched difference over these bins (own minus others') is +24.1 [12.7, 39.9].

Changed lines	Own failing (n)	Others' failing (n)	Leniency, own	Leniency, others'
1–5	29	19	+14.2	−7.1
6–15	32	13	+11.2	−17.1
16–40	11	3	+2.6	−11.6
41 or more	5	5	+29.1	−5.6

E Models judging their own patches

On the Mini issues, five of the other writers judged their own mini-SWE-agent patches, so the harness is matched for them by construction. None shows a gap (Table 4). The pooled gap over the five is −0.036 [−0.089, 0.016].

We also measured their leniency on their own failing patches (Table 5). The intervals come from the same issue resamples as the gap. We computed them on 2026-10-22, after the point estimates had been seen, under a plan that counts a model as lenient only if its interval stays above zero after a Bonferroni correction for five models. Each model wrote only 14 to 24 failing patches, so the intervals are wide. The pooled leniency difference over the five is +5.1 [−0.8, 11.0] points. Gemma's is +20.9 [11.5, 30.2] on the same issues. That is 15.8 [4.0, 27.7] points above the pooled five. By the planned rule, none of the five is shown to be lenient.

Llama-4-Maverick comes closest. Its leniency difference is +17.5 [0.6, 32.9], and its Bonferroni interval [−3.5, 38.4] includes zero. Maverick gives high scores to most failing patches, whoever wrote them. It scores 95 or above on 69% of other writers' failing patches and on 57% of its own. So it does not single out its own patches the way Gemma does.

On the held-out issues, all nine judges also scored the five writers' own patches, which all come from mini-SWE-agent (Table 6). Each writer has 46 to 123 failing patches there, three to six times as many as on the Mini issues. We chose this analysis on 2026-10-22, after the Mini results above and a review of the paper (`spec/other-models-heldout.md`). The code is the Mini code unchanged, and it reproduces Gemma's held-out gap and leniency in the paper (Sections 3.1 and 3.3). The pooled gap over the five is −0.001 [−0.031, 0.029], and the pooled leniency difference is +0.4 [−2.5, 3.4] points. Gemma exceeds the pooled five by 0.075 [0.015, 0.133] in gap and by 16.3 [8.9, 23.4] points in leniency. By the same Bonferroni rule, none of the five is shown to have a gap or leniency. Llama-4-Maverick's leniency difference falls to +2.8 [−5.0, 10.8], and it gives 100 to fewer of its own failing patches than of other writers' (33% against 42%).

Table 4. Models judging their own patches on the Mini issues (all patches from mini-SWE-agent).

Model	AUROC, own patches	AUROC, other families' patches	Self-judging gap [95%]
Qwen3-Coder	0.691	0.786	−0.032 [−0.241, 0.189]
gpt-oss-120b	0.860	0.701	−0.046 [−0.133, 0.049]
GLM-4.5	0.757	0.752	−0.047 [−0.174, 0.116]
Kimi-K2	0.881	0.805	−0.036 [−0.113, 0.036]
Llama-4-Maverick	0.786	0.657	−0.019 [−0.133, 0.101]

Table 5. Leniency difference of models judging their own patches on the Mini issues, in points: mean leniency on own patches minus mean leniency on other writers' patches. Gemma's patches come from four harnesses, the others' from mini-SWE-agent.

Model	Own failing	Failing patches [95%]	Failing patches [Bonferroni 99%]	Passing patches [95%]	Failing scored 100: own / others'
Qwen3-Coder	15	+4.1 [−6.1, 14.6]	[−8.2, 18.3]	+3.5 [−15.5, 20.8]	7% / 0%
gpt-oss-120b	24	+2.1 [−9.1, 13.0]	[−11.3, 15.6]	+12.1 [5.1, 19.9]	0% / 0%
GLM-4.5	14	+7.1 [−10.5, 26.3]	[−17.7, 31.6]	+4.4 [−11.4, 17.5]	7% / 16%
Kimi-K2	14	−5.4 [−17.6, 7.2]	[−22.1, 11.6]	+4.9 [−2.7, 12.4]	0% / 2%
Llama-4-Maverick	21	+17.5 [0.6, 32.9]	[−3.5, 38.4]	+7.9 [−3.7, 19.6]	43% / 40%
Pooled, five models		+5.1 [−0.8, 11.0]			
Gemma 4 31B	80	+20.9 [11.5, 30.2]	[8.4, 32.6]	+0.2 [−7.6, 8.7]	61% / 20%

Table 6. Models judging their own patches on the held-out issues (all patches from mini-SWE-agent; exploratory). Leniency is the difference on failing patches, in points.

Model	Own failing	Self-judging gap [95%]	Leniency, failing patches [95%]	Leniency, failing patches [Bonferroni 99%]	Failing scored 100: own / others'
Qwen3-Coder	46	−0.040 [−0.100, 0.023]	−8.1 [−15.4, −0.5]	[−17.3, 1.9]	0% / 0%
gpt-oss-120b	109	+0.001 [−0.066, 0.073]	+2.9 [−3.3, 8.8]	[−5.0, 10.8]	0% / 0%
GLM-4.5	61	−0.048 [−0.113, 0.019]	+0.8 [−8.8, 9.7]	[−11.8, 12.3]	12% / 9%
Kimi-K2	54	+0.035 [−0.035, 0.103]	+3.5 [−3.9, 10.6]	[−6.2, 13.2]	4% / 1%
Llama-4-Maverick	123	+0.048 [−0.021, 0.117]	+2.8 [−5.0, 10.8]	[−7.7, 12.5]	33% / 42%
Pooled, five models		−0.001 [−0.031, 0.029]	+0.4 [−2.5, 3.4]		
Gemma 4 31B	85	+0.074 [0.022, 0.126]	+16.6 [10.3, 23.1]	[7.6, 24.5]	61% / 35%

F Per-judge results on the held-out issues

Table 7. Held-out issues, per judge: share of failing patches scored 100 on Gemma's patches and on the other writers' patches; AUROC on Gemma's patches; and, for outside judges, the drop from their AUROC on other writers' patches to their AUROC on Gemma's (the mean drop, 0.055, enters the self-judging gap).

Judge	Gemma's failing at 100	Others' failing at 100	AUROC, Gemma's	Drop
Gemma 4 31B	61.2%	34.6%	0.622	
Llama-4-Maverick	45.9%	38.7%	0.611	0.059
GLM-4.7-Flash	36.5%	24.9%	0.636	0.091
Qwen3.5-9B	21.2%	23.4%	0.685	0.050
GLM-4.5	10.6%	9.2%	0.778	0.053
Kimi-K2	1.2%	1.8%	0.766	0.055
gpt-oss-20b	0%	0%	0.665	0.038
gpt-oss-120b	0%	0%	0.656	0.074
Qwen3-Coder	0%	0%	0.717	0.022

G Checks of Gemma's patches

Table 8. AUROC on Gemma's own patches minus Gemma's AUROC on the same patches, with 95% and Bonferroni (eight judges) intervals. Exploratory.

Check	Held-out [95%]	Held-out [Bonferroni]	Mini, mixed / matched
GLM-4.5	+0.156 [0.083, 0.228]	[0.054, 0.256]	+0.082 / +0.103
Kimi-K2	+0.144 [0.081, 0.205]	[0.061, 0.230]	+0.129 / +0.037
Mean of 8 outside judges	+0.129 [0.067, 0.194]	[0.039, 0.216]	+0.130 / +0.078
Qwen3-Coder	+0.095 [0.025, 0.163]	[−0.004, 0.187]	+0.098 / −0.045
Qwen3.5-9B	+0.063 [−0.007, 0.138]	[−0.038, 0.163]	+0.127 / −0.007
gpt-oss-20b	+0.043 [−0.035, 0.128]	[−0.067, 0.156]	+0.032 / −0.006
gpt-oss-120b	+0.034 [−0.036, 0.106]	[−0.064, 0.131]	+0.059 / +0.101
GLM-4.7-Flash	+0.014 [−0.066, 0.090]	[−0.099, 0.118]	+0.064 / −0.125
Llama-4-Maverick	−0.011 [−0.083, 0.060]	[−0.116, 0.092]	−0.032 / −0.050
Gemma + GLM-4.5	+0.146 [0.094, 0.197]	[0.074, 0.213]	+0.083 / +0.098
Gemma + Kimi-K2	+0.134 [0.088, 0.181]	[0.070, 0.196]	+0.114 / +0.034
Gemma + Qwen3-Coder	+0.101 [0.053, 0.151]	[0.030, 0.169]	+0.114 / +0.053
Gemma + Qwen3.5-9B	+0.081 [0.032, 0.136]	[0.011, 0.154]	+0.119 / +0.037

H Submitting without review: all rates

Table 9. Share of submitted patches that pass, by acceptance rate and check. Gemma's own cut (every patch it scores 100) accepts 73.9% of held-out patches, 76.2% of matched Mini patches and 73.3% of mixed Mini patches, with 60.9%, 62.5% and 55.5% passing. Base rates: 52.8%, 50.0%, 46.7%.

Patch set	Check	Top 25%	Top 50%	Top 75%
Held-out (180)	Gemma	60.9%	60.9%	61.0%
	Gemma + Qwen3.5-9B	75.8%	67.5%	61.1%
	Kimi-K2	77.0%	72.4%	63.7%
	GLM-4.5	83.6%	73.5%	63.0%
Mini, matched (42)	Gemma	62.5%	62.5%	62.5%
	Gemma + Qwen3.5-9B	70.0%	68.5%	61.9%
	Kimi-K2	65.6%	67.4%	61.5%
	GLM-4.5	70.6%	71.4%	66.7%
Mini, mixed (150)	Gemma	55.5%	55.5%	54.7%
	Gemma + Qwen3.5-9B	71.2%	64.7%	56.6%
	Kimi-K2	68.8%	67.9%	57.8%
	GLM-4.5	68.3%	62.4%	56.9%

The paper's table shows three second-judge checks, chosen after their AUROCs were seen. Table 10 gives all 16 checks we ran on the held-out patches. In the top quarter, the median gain over Gemma is +11.7 points, and 11 of the 16 intervals lie above zero. In the top half, the median is +6.2 points, with 8 of 16 above zero.

Table 10. Every second-judge check on Gemma's 180 held-out patches: share passing in the submitted top quarter and top half minus Gemma's, in points, with 95% intervals from 2,000 resamples of issues. "Gemma + X" averages the two judges' ranks. `derived/tie_break_2026-10-22.json`.

Check	Top quarter	Top half
Gemma + GLM-4.5	+25.1 [14.8, 33.8]	+11.6 [6.0, 16.6]
GLM-4.5	+22.7 [12.7, 32.0]	+12.6 [6.8, 18.3]
Gemma + Kimi-K2	+16.7 [10.7, 22.9]	+12.2 [6.9, 17.7]
Kimi-K2	+16.1 [9.6, 22.5]	+11.5 [5.9, 17.6]
Gemma + Qwen3.5-9B	+14.9 [6.7, 23.3]	+6.6 [1.5, 12.0]
Qwen3.5-9B	+13.7 [5.7, 22.2]	+7.3 [1.6, 13.5]
Gemma + Qwen3-Coder	+13.3 [6.4, 20.6]	+9.9 [4.2, 15.6]
Qwen3-Coder	+12.6 [5.5, 20.0]	+11.1 [4.1, 17.0]
Gemma + gpt-oss-20b	+10.8 [3.0, 18.3]	+3.9 [−2.3, 9.7]
gpt-oss-20b	+9.6 [1.6, 17.5]	+4.3 [−2.5, 11.4]
Gemma + GLM-4.7-Flash	+9.0 [2.9, 16.0]	+4.7 [−0.7, 12.0]
gpt-oss-120b	+8.1 [−1.6, 17.3]	+3.5 [−2.1, 10.8]
Gemma + gpt-oss-120b	+7.5 [−2.2, 17.0]	+3.5 [−2.1, 9.8]
GLM-4.7-Flash	+5.8 [−1.3, 13.0]	+5.8 [−2.0, 12.1]
Gemma + Llama-4-Maverick	+3.5 [−2.4, 9.6]	+3.0 [−2.7, 8.5]
Llama-4-Maverick	+0.1 [−6.2, 6.6]	+0.1 [−6.2, 6.6]
Median of the 16	+11.7	+6.2

Gemma's own signals

Breaking the tie at 100. We fixed three tie-breaks on the Mini issues before computing any held-out value (Table 11; `spec/tie-break.md`). Inside Gemma's score, each ranks one kind of patch higher: fewer concern words in Gemma's answer, a shorter answer, or fewer model calls in Gemma's agent run. Only our own runs record model calls, so that tie-break was chosen on the 32 matched-harness Mini patches Gemma scored 100. On the held-out issues, concern words do not help, a shorter answer helps a little, and fewer model calls help most. The rule "Gemma scores 100 and the run used at most 9 model calls" submits 41 held-out patches (22.8%). Of these, 78.0% pass. Inside the tie, the number of model calls correlates with the outside judges' mean score (Spearman −0.57), so run length may partly stand for how hard the issue is. Against this tie-break (exploratory), GLM-4.5's top quarter passes +5.9 points more [−3.6, 16.4] and its top half +1.9 [−5.2, 10.4]. Kimi-K2's differ by −0.8 [−10.7, 10.7] and +0.7 [−6.6, 9.1].

Table 11. Tie-breaks inside Gemma's score, fixed on the Mini issues before the held-out test. AUROC inside the tie separates passing from failing patches among those Gemma scored 100. Gains are share passing minus Gemma's, in points, with 95% intervals. `derived/tie_break_2026-10-22.json`.

Tie-break	AUROC in tie, Mini	AUROC in tie, held-out	Top quarter	Top half
Fewer concern words	0.527 (142)	0.533 [0.438, 0.630]	+6.1 [−4.2, 16.8]	0.0 [−5.0, 5.5]
Shorter answer	0.621 (142)	0.626 [0.529, 0.717]	+10.9 [0.3, 21.7]	+4.1 [−1.4, 10.5]
Fewer model calls	0.731 (32)	0.679 [0.578, 0.769]	+16.9 [5.7, 26.8]	+10.8 [4.2, 16.5]

Table 12. Signals that need no second model, on 172 of Gemma's mixed-harness patches with both Gemma scores (planned before computing; `spec/self-check-fix.md`). The planned test was the score plus attempt agreement (mean overlap of changed lines with Gemma's other attempts at the issue); its bar was a gain of at least 0.05 with a lower bound above zero.

Signal	AUROC	Change from Gemma's score [95%]
Gemma's score, reasoning off	0.646	
Gemma's score, thinking on	0.645	−0.001 [−0.056, 0.058]
Mean of the two scores	0.677	+0.031 [−0.005, 0.069]
Attempt agreement	0.665	+0.019 [−0.119, 0.146]
Score plus attempt agreement (planned)	0.720	+0.074 [−0.014, 0.160]
Score relative to its other attempts	0.533	−0.113 [−0.197, −0.025]

J Cost

Table 13. API cost as reported by OpenRouter per call. gpt-oss-120b calls were reported at zero cost by their provider. Gemma's 250 agent runs took 49 agent-hours on one NVIDIA L40S of a university cluster, at no charge.

Run	Calls	Cost (USD)
Four judges on Gemma's own attempts (ratings reused)	3,244	0.53
Nine judges, Mini issues, mixed-harness patches	3,292	2.66
Gemma 4 31B with thinking on, Mini issues	401	0.51
Nine judges, held-out and matched-harness patches	9,857	8.66
Total	16,794	12.36